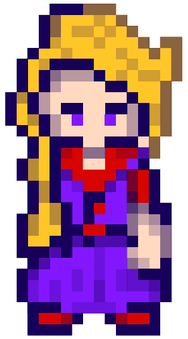

CIRCULAR
SILO
Developer: Deepthi Nenavath (dn24002)
PROTOTYPE I
A REPAIR-AND-RECOVERY GAME ABOUT RESOURCE WASTE
PYTHON AND PYGAME, PC

GAME DESCRIPTION



- SINGLE-PLAYER RESOURCE MANAGEMENT
- UNDERGROUND SILO SURVIVAL
- TOXIC SURFACE, SCARCE RESOURCES



- REPAIR · REUSE · RECOVER · INCINERATE
- MENTORED BY MARTHA
- BEFORE TIMES RELICS

IMPLEMENTED MECHANICS (LEVELS 1 TO 3)

Progression: Each level needs 100 points, and a score that falls too low sends you back a level.



Level 1, Sorting: Objects arrive on a conveyor belt. You inspect each one and choose Reuse, Repair, Recover or Incinerate. The score is +10 for a correct choice and -5 for a wrong one.



Level 2, Material Recovery: You take apart the objects you sent to Recover, pick the usable parts and store them. Each usable part is +10 and each damaged part picked is -5.

Level 3, Learning to Repair: Martha explains each repair step by step, and you pick the matching action from six options. A correct action is +5 and a wrong one is -5. A finished repair is +15.

WORKING FEATURES

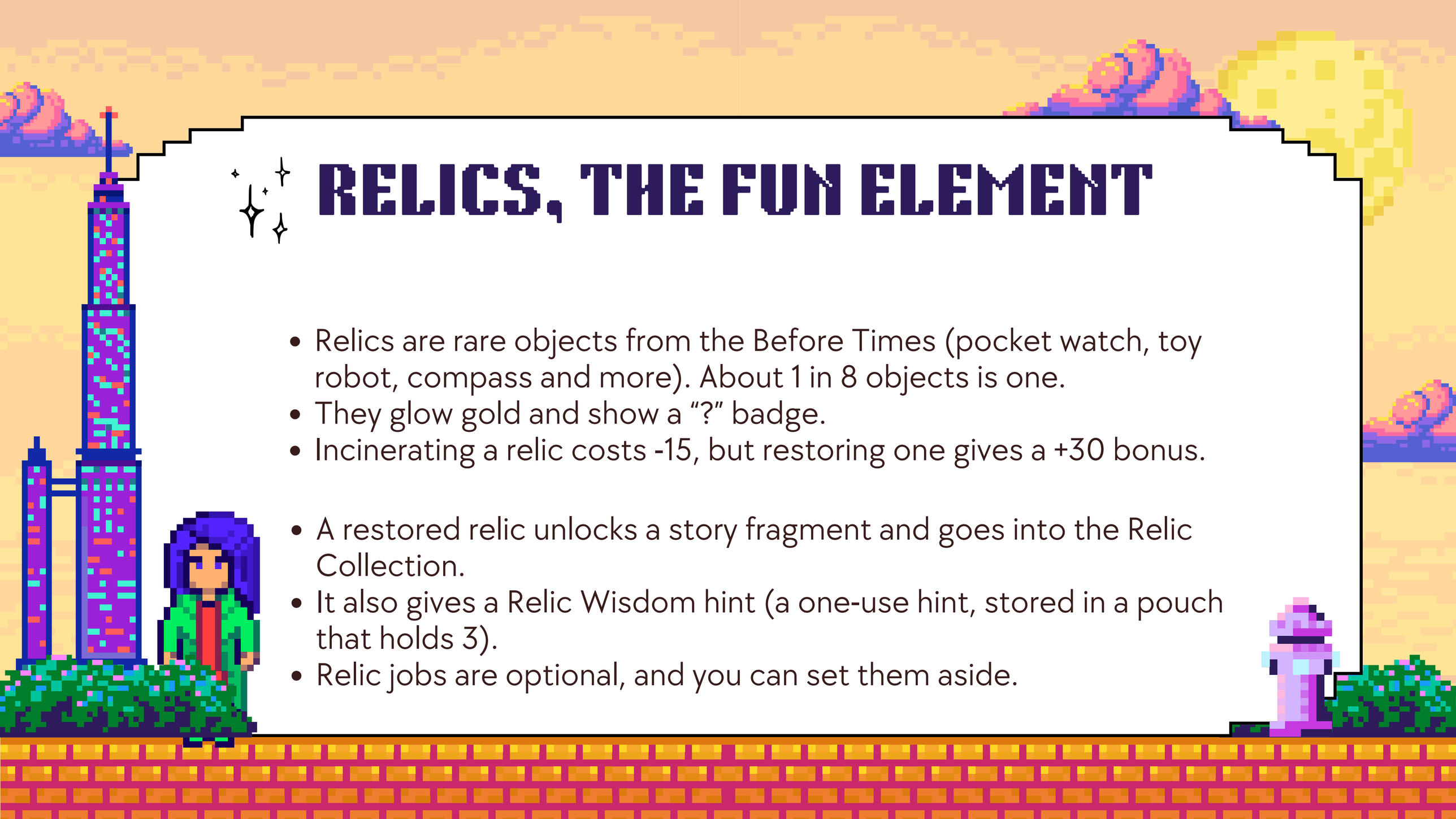
- Main menu with a live score and level display
- 20 ordinary objects and 8 relics, each with inspection clues
- Feedback after every decision that explains why
- Parts storage shared between levels
- 13 repair jobs with Martha's instructions
- Replacement parts taken from your Level 2 storage
- Progress bar, completion screens and a drop-back screen
- Mouse and keyboard controls
- The whole game flows from the menu through Level 3



RELICS, THE FUN ELEMENT

- Relics are rare objects from the Before Times (pocket watch, toy robot, compass and more). About 1 in 8 objects is one.
- They glow gold and show a “?” badge.
- Incinerating a relic costs -15, but restoring one gives a +30 bonus.

- A restored relic unlocks a story fragment and goes into the Relic Collection.
- It also gives a Relic Wisdom hint (a one-use hint, stored in a pouch that holds 3).
- Relic jobs are optional, and you can set them aside.



SOCIAL GOOD INTEGRATION (SDG 12)

Issue: Resource waste.
Usable products and parts are discarded too easily.

Goal: SDG 12, Responsible Consumption and Production.

Cycle: Sort → Recover → Repair → Reuse

In the gameplay now:

- Throwing away something repairable costs you points.
- Recovered parts become real replacement parts in Level 3.
- Relics show that some objects are worth keeping for what they mean.



Message: A broken object is not automatically waste.

PROTOTYPE II

- **Level 4** — Guided Repair: Score repairs based on the percentage of correct steps, with Martha offering only general advice.
- **Level 5** — Independent Repair: Repair without Martha's help, relying on recovered parts and relic hints.

LEVEL 4: ADVANCED REPAIR

- Martha gives only general advice, so the player has to work out the repair alone.
- The player inspects the object and marks the components that look faulty.
- From 8 possible actions (5 needed, 3 wrong), the player picks the right ones and puts them in order.





- The player presses Done Fixing and Martha reviews the repair. She shows the correct order next to the player's order.
- A restored relic gives +30 and a Relic Wisdom hint. The player can use these hints here.
- Falling below 300 points sends the player back to Level 3.

LEVEL 5: INDEPENDENT REPAIR

- Martha gives no advice and no review, so the player applies what Levels 3 and 4 taught.
- The player only sees whether the repair holds, plus the points. The correct answer is never revealed.





- The only help is Relic Wisdom from the Hint Pouch.
- Scoring is the same as Level 4. Falling below 400 sends the player back to Level 4.
- Reaching 500 points finishes the game and shows an ending screen.

PLANS FOR FINAL

- **Relic Hint Effects**: Give relics unique abilities, such as a pocket watch revealing step order or a camera exposing defects.
- **Relic Collection**: Add a screen to view collected relics and their discoveries.

AI USAGE

Claude

- **Claude (Anthropic):** Brainstorming, README writing, code development and refactoring, and game content.

- **Google Gemini:** Generated pixel-art images.



- **My Contribution:** Wrote parts of the code, reviewed and tested AI-assisted code, removed redundancy, and made the final design, story, and gameplay decisions.

THANK YOU

Individual project. Concept, design, README, programming, content and testing by Deepthi Nenavath

